
Reliability of sequence-alignment analysis of social processes: Monte Carlo tests of ClustalG software

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Abstract. Sequences of characters are used in many fields to record events or processes that characterize social processes. However, until recently, there have been very few methods available for the analysis of character-sequence data. Alignment algorithms measure similarities between pairs of sequences by inserting gaps into one or the other to create the best possible matching pattern. In this paper the reliability of alignments in the classification of sequential data is examined. Alignment methods were developed in computational biology, but are being considered for applications in other fields such as sociology, geography, and transportation planning. The ClustalG multiple alignment package is used to examine a set of synthetic sequences generated through the use of eight separate generation rules. Through the application of the software to sequential data with a known number of subgroups and known patterns in the sequences, some strategies for conducting the analysis can be compared and evaluated. The most effective strategy for analysing sequential data when the underlying processes that generate the event sequences are not known is to use low gap penalties that permit the maximum numbers of matches.

1 Introduction

In this paper are described a number of tests of alternative strategies for the alignment analysis of sequential data using the ClustalG software package. Alignment algorithms, which were originally developed for use in biochemistry, calculate similarity or distance measures for pairs of character sequences according to various parameter settings. Different strategies use different parameter sets. These are applied to character sequences produced by a Monte Carlo generator to test the reliability of alignment methods in finding known patterns in a synthetic dataset. Success in finding known patterns is interpreted as indicating that alignment methods are reliable, and the relative effectiveness of various parameter-setting strategies illustrates the relative reliability of each strategy. An alternative sequence-similarity measure is used to compare similarity patterns identified using alignment measures.

Alignment algorithms were introduced to social science first by Abbott in sociological applications in the mid-1980s. In his 1995 paper he discussed the advantages of the examination of social processes from a sequential perspective and advocated the application of alignment methods in social research (Abbott, 1995). Alignment algorithms base the similarity of sequence pairs on the frequency and position of characters and on the pattern of character transitions. They are thus more powerful and flexible than alternative sequence-comparison techniques, such as dot matrices, or simple match comparisons.

Abbott and Tsay (2000) is the most thorough available review of applications in sociology, political science, history, and geography. A number of sociologists have used alignment methods in studies of job promotions and career development (Blair-Loy, 1999; Stovel et al, 1996; Williams and Han, 2003). In transportation research the daily activities of individuals and households have long been hypothesized to be the key determinants of travel demand (Chapin, 1974; Jones, 1990; Pas, 1982). Alignment analysis of

daily time use was illustrated in Wilson (1998a) and analyses of the activity–travel relationship are now appearing (Schlich and Axhausen, 2003; Wilson, 1998b). Renaud (2001) used ClustalG in his study of the behaviour of immigrants to Quebec. In the area of transportation planning, Joh et al (2001a; 2001b) have developed several variants of the basic alignment algorithms for testing models of travel behaviour. Joh's DANA (Dissimilarity Analysis of Activity-travel patterns) software package implements these and a number of other analytic procedures for comparing sequential data.

Any mathematical procedure will produce some result if data are supplied as required. Researchers using relatively new procedures have to assure themselves and the wider research community that the results are valid and reliable, and that such results are not generated by the methods themselves. Sequence-alignment methods should be handled with particular care because they are at present relatively unknown to social researchers. Levine (2000), in a critique of Abbott and Tsay, raised the issue of the validity of alignment results. Consequently, though interesting and suggestive results have been reported, it is important to address the question of the reliability and the validity of the methods.

Social researchers frequently face problems in measuring abstract concepts and there is a large literature that examines the validity and reliability of measurement methods. To quote from Carmines and Zeller (1979, page 16), "Reliability is basically an empirical issue, focusing on the performance of empirical measures Validity ... is evidenced in the degree that a particular indicator measures what it is supposed to measure rather than reflecting some other phenomenon." Concerns regarding the use of alignment algorithms and the accuracy of results based on alternative sets of parameter settings for alignment software involve the *reliability* of alignment methods. Discussions of the *validity* of alignment methods relate to the fidelity of, for example, diary data as a medium that conveys information about daily activity. Given that data on activity, migration, careers, or whatever, have been collected and have been recorded as character sequences, the validity bridge has been crossed. Researchers should rightly be concerned that alignment methods in general and parameter settings in particular identify similar individuals reliably. An examination of this question is the objective here.

2 Sequence comparison methods

2.1 Sequence alignment

2.1.1 Pairwise alignments

The writing of a pair of sequences, one above another, creates some degree of alignment between sequence elements. Finding an optimal alignment involves transforming elements of one sequence into elements of the other using a defined set of operations. Optimality of alignment means that no alternative arrangement of matching characters and inserted gaps can give a higher similarity (or lower distance) score than the one found using only eligible operations.

Eligible operations in most pairwise alignment algorithms are identities (or exact matches), substitutions (or inexact matches), the insertion of an element from one sequence into the other, and the converse operation of deletion of an element. Gaps are created in either sequence, as necessary, to accommodate insertions and deletions. Insertions and deletions are descriptions of the same operation from the perspective of one or other sequence in the pair. They always occur in pairs and are usually called indels. Alignment algorithms can be designed to calculate either distances between sequences or similarities. In this paper the discussion will be mainly in terms of similarities but all comments apply equally to analyses based on distances.

Examples of two possible alignments of the sequences [ties] and [untied] are illustrated below. To determine which is superior, one would have to define the relative weights of substitutions and indel penalties.

--ties --tie-s
untied untied-

If whole sequences are examined the method is called *global* alignment. A longest common subsequence may also be found using *local* alignment algorithms. ClustalG offers both approaches to pairwise alignments. Abbott (1995), Durbin et al (1998), Harvey and Wilson (2001), Joh et al (2001b), Waterman (1995), and Wilson (2001) give formal accounts of alignment algorithms. Informally, the algorithms quantify the visual pattern of matches in a comparison table as shown in figure 1. The ‘1’ in a cell indicates a match. Two identical sequences would produce a square table that matched perfectly on the principal diagonal. Generally, a strong diagonal pattern reflects strong sequence similarity.

Researchers in the social sciences tend to use pairwise similarity matrices as inputs to some type of classificatory method such as clustering or principal components analysis. Biochemists more often create multiple alignments based on pairwise similarities.

	u	n	t	i	e	d
t			1			
i				1		
e					1	
s						

Figure 1. A sequence comparison table.

2.1.2 Multiple alignment

Pairwise alignment algorithms may be generalized to any number of sequences, say, *m*. However, the alignment of *m* sequences causes an explosive growth of the computational problem. Exact methods can handle only about ten sequences, depending on length. Most multiple-alignment software, including ClustalG, employs approximate methods to construct alignments of many sequences.

Multiple alignments are a graphic representation of the input sequences, adjusted to accommodate gaps. They can also be used to generate a revised similarity matrix. However, as the majority of existing applications of alignment methods to social problems use pairwise distance or similarity matrices, the properties of the multiple-alignment results will not be examined in this paper.

2.1.3 Indel and matching penalties

ClustalG gives a similarity value of 10 for matches, 0 for mismatches, if an identity matrix is used, and uses separate opening and extension penalties for indels. High values of the gap penalties compared with matching scores will compress the alignment and, in the limit, will suppress any gapping at all, reducing the analysis to a count of matches. Very low gap penalties allow the maximum number of identical matches, but the alignments are highly spread and bear little resemblance to the raw sequences. Gap penalties are often set to be at least half the value of the average score for a match so as to force the alignment to achieve a significant gain in exchange for the displacement introduced by the gap. Values of 1.0 for the opening penalty and 0.1 for the

extension penalty are used as the low-gap options and values of 8 and 3 are used for the high-gap options. Alignments based on something such as the high gap penalties make the analysis of time-use data look realistic, meaning that the major daily activities, such as work, sleep, eating meals, and recreation, are present approximately at their relative times during the day.

There have been many efforts to derive relative similarity measures for event substitutions in social research. One of the most common strategies has been to examine transition probabilities among events. The argument is that events with low transition probabilities are unrelated and their substitution should be penalized more severely than mismatches among events that are related in the form of event chains. Other strategies involve consideration of key dimensions of events on the basis of the substance of the research. For example, in career studies, occupations with different training requirements or wage rates could be considered to be substantially different, and substituting one for another would be penalized more severely than substitutions among similar occupations. This paper is concentrated on the analysis of indel penalties, and substitutions or mismatches are given zero similarity.

2.2 Gower (Euclidean) sequence similarity

The Gower or Euclidean similarity (Gower, 1971; Koppelman and Pas, 1985) was designed to measure the similarity of two objects consisting of k attributes. The measure assumes that objects have the same number of attributes. The similarity of objects i and j is defined as

$$g_{i,j} = \sum_k w_k s_{i,j,k} / \sum_k w_k,$$

where w_k is the weight of the attribute k , and $s_{i,j,k}$ is the similarity of objects i and j for attribute k . Let the attributes, k , indicate the position of an element in the sequence. If the elements of sequence i and j match at position k , then $s_{i,j,k} = 1$. Otherwise $s_{i,j,k} = 0$.

The relative importance of different attributes is measured by weights. However, in cases in which k indicates the sequence position and no other attribute is involved, all weights are 1. The measure is then simply a ratio of the number of observed identities to the maximum number possible. Because all sequences have the length 50 in this experiment, the measure further simplifies to the number of matches between two sequences. The general approach is called Euclidian because a squared distance measure can be defined in terms of the similarity terms.

The Gower measure can be interpreted as an extreme case of alignment in which all indels are suppressed by high penalties, leaving only exact matches as positive scores. Tests of the global algorithm that set the gap-opening penalty to 100 gave exactly the same scores as the Gower measure. If sequences are very similar, we may expect to find that the Gower measure performs fairly well in comparison with alignments because the probability of a match at any position is high. Alignment measures should be relatively better for fuzzier sequence-generation processes, which present more difficult classification problems.

Pas (1982) and Koppelman and Pas (1985) developed extensions to the Gower measure for transportation analysis in order to accommodate multiple attributes of travel activity such as purpose, mode, and other key travel-related variables. The Koppelman–Pas measure may also have an interpretation as a special case of an alignment.

2.3 Character counting

Another way of comparing sequences has been suggested that is an important alternative to intersequence similarity matrices. One can simply count the number of characters of each type, thereby creating a vector of frequencies that adds up to the sequence length. This approach ignores the sequential arrangement of elements and involves only counts of elements. It is analogous to the time-budget representation of daily activities, which adds up episode durations, ignores their timing and transition, and reports only total time spent according to activity category. As another example, if the research were to involve employment or migration events, the counts would simply be the duration of events of each type that individuals experienced. At face value this would seem to be a poor representation of the information in sequential data, but its simplicity and ease of use demand that its reliability be tested also. No interitem similarity measures are necessary because each item already consists of numerical measurements, and clustering methods can be applied directly to the input data. The clustering accuracy of the character-counting approach is reported in section 5.

3 Software and data

3.1 ClustalG multiple alignment package

ClustalG is a version of ClustalW, which was developed at the European Molecular Biology Laboratory (Thompson et al, 1994), amended for use outside of biochemistry. ClustalG is available from www.stmarys.ca/partners/iatur/clustalg. ClustalG has deleted the explicitly biochemical features of its parent packages. It has also expanded the input routines to accept multiple-letter words of up to twelve characters, and has created new diagnostic files for analysis of the clustering process. The key feature is the introduction of multiple-letter words permitting analysts to use complex coding schemes where necessary. Analysts may use different positions in the word to indicate different dimensions of events. It should be noted that Joh et al (2001a) have approached the analysis of multidimensionality directly by developing algorithms that align multiple sequences for each sample member.

3.2 Pairwise similarities and distances

ClustalG writes pairwise similarity values to an output file. Gower similarity measures were calculated from input sequences by the Statistical Analysis System (SAS). All similarity matrices were transformed into distance matrices and the Ward clustering algorithm was then used—implemented in the SAS CLUSTER procedure to compare how well different measurement approaches identified known patterns in the test sequences.

3.3 Monte Carlo sequence generation

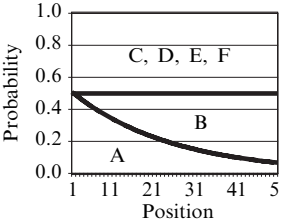
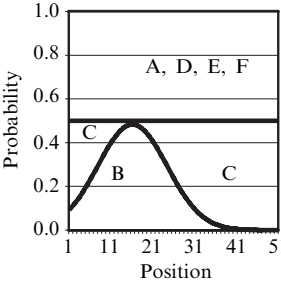
The Monte Carlo simulator written for this project produced sequences of the six letters, A to F, in a format known as 'Pearson' or 'Fasta'. Each sequence record begins with the 'greater than' sign followed by identification fields. The character sequence proper begins in the first position of the second line and occupies as many positions as are needed. The user determines the number and length of sequences to be generated. The Courier font is convenient for illustrating sequence data because all characters are the same width. An example of the format is the following worker's day, in which the major activities are: sleep, z; work, w; television, m; and travel, t.

```
>1633tha 15e 51
```

```
zzzzzzptwwwwwwwwewwwwwwwwtsssteemmmmmmpmmmmzzzzzz
```

The patterns in the Monte Carlo sequences are produced by eight generation rules. The generators define patterns on the basis of negative exponential probabilities to the first and second powers and uniform probability distributions. The probability of

Table 1. Eight Monte Carlo sequence generation rules.

Rule identifier	Function	Description	Probability space
ABnex	negative exponential	‘A’ located in the initial positions and ‘B’ located towards the right end of the sequence; other characters are uniformly located across the sequence. pattern letters: A, B residual: C, D, E, F	 $P = \text{PROB} \exp(-0.046k)$
BAnex	negative exponential	‘B’ and ‘A’ reversed from ABnex. pattern letters: B, A residual: C, D, E, F	As ABnex
EFnex	negative exponential	‘E’ replaces ‘A’, ‘F’ replaces ‘B’ in ABnex. pattern letters: E, F residual: A, B, C, D	As ABnex
FEnex	negative exponential	‘F’ replaces ‘A’, ‘E’ replaces ‘B’ in ABnex. pattern letters: E, F residual: A, B, C, D	As ABnex
BCnx2	second-order negative exponential	‘C’ is concentrated in left and right positions. ‘B’ located in left centre. pattern letters: B, C residual: A, D, E, F	 $P = 0.2\text{PROB} \times \exp(0.21k - 0.007k^2)$
DEnx2	second-order negative exponential	‘D’ replaces ‘B’, ‘E’ replaces ‘C’ in BCnx2 pattern letters: D, E residual: A, B, C, F	As BCnx2
ABC75	uniform	majority of letters A, B, and C but no left–right trend. PROB not used.	probability (A or B or C) = 0.75 probability (D or E or F) = 0.25
DEF75	uniform	majority of letters D, E, and F but no left–right trend. PROB not used.	probability (D or E or F) = 0.75 probability (A or B or C) = 0.25

the occurrence of a patterned character is a function of its position, k , in the sequence. The user may specify the number and length of sequences and may choose characters or numerals. A parameter called PROB varies from 0 to 1 and controls the portion of the probability space that is used for patterned characters, as opposed to the

proportion used for residual or ‘noisy’ characters. Residual characters are equally probable in the space not occupied by patterned characters. Higher values of PROB reduce noise and create clearer patterns. Table 1 defines the generation rules and shows the probability spaces for patterned characters and for the residual characters.

Sequences generated by the same rule will all be different because the selection of a letter at each position is probabilistic. However, sequences will share characteristics written into the generation rules. Sequences generated by different rules should be recognizably different but, given the probabilistic nature of the processes, it is possible for individual sequences to resemble those made by different rules. Figure A1 in the appendix shows two examples of each of the eight sequence types for PROB set to 0.80. The sequences are highly distinct and identification of the generators is left as an exercise for interested readers. Identification of examples from datasets with PROB set to 0.50 or less is considerably more difficult.

The eight generation rules illustrate three kinds of processes. The four rules in table 1 based on the first-order negative-exponential function implement a process that shows a change from an initial state to a subsequent state. The two rules based on the second-order function show a process that evolves from one state to a second then back to the first. The last two rules, based on rectangular distributions, simply show a more frequent set of states and a less frequent set.

The one-way process can be interpreted as a model of evolutionary or staged change and could be applied to any of a wide range of research fields that examine the introduction of technical or social innovations. It may be a short-term model in the sense that only two stages, of possibly many, are examined. The second-order function could be interpreted as a model of a process that begins slowly, grows to a peak, then passes. Daily activity diaries illustrate a situation in which a set of rest or personal-care activities in the morning gives way to work and social activities, then reoccurs at the end of the day. The emergence and passing of fads is another example. The second-order process may be thought of as modeling the rate of change of a logistic process.

3.4 Performance of the generator

Table 2 shows the probabilities of each character expected in a fifty-character sequence with PROB set to 0.5 and the default exponential parameters. The residual characters in the exponential generation rules will always have theoretical probabilities of one quarter of the value (1 – PROB), or 12.5% when PROB is 0.5. The probabilities of the characters set by the exponential functions are derived by integrating the functions from 0 to 50. A set of 200 sequences creates 10 000 characters. χ^2 tests of the actual character counts indicate that frequencies are not significantly different from those expected. Note that the character-count data files consist of records containing counts which resemble the rows of table 2.

Table 2. Probabilities of characters generated by eight rules for fifty-character sequences.

Generation rule	Expected percentage of characters					
	A	B	C	D	E	F
ABC75	25.0	25.0	25.0	8.0	8.0	9.0
DEF75	8.0	8.0	9.0	25.0	25.0	25.0
ABnex	21.5	29.5	12.5	12.5	12.5	12.5
BAnex	29.5	21.5	12.5	12.5	12.5	12.5
EFnex	12.5	12.5	12.5	12.5	21.5	29.5
FEnex	12.5	12.5	12.5	12.5	29.5	21.5
BCnx2	12.5	19.8	30.2	12.5	12.5	12.5
DEnx2	12.5	12.5	12.5	19.8	30.2	12.5

3.5 Test data

Sequence similarity was measured for sets of 240 sequences composed of thirty sequences generated under each of the eight rules. Each experiment employed three sequence sets using different levels of the PROB parameter. The PROB80 set, based on a PROB value of 0.80, distinguishes very clearly between the eight generation rules. The PROB50 and PROB35 sets, based on PROB values of 0.50 and 0.35, respectively, present progressively more difficult classification problems. The experiment was replicated ten times.

All sets were aligned with ClustalG to calculate similarity matrices using the low indel penalties and the local and global pairwise algorithms (designated L1 and G1) and using the high penalties and both algorithms (designated L8 and G8). The Gower similarity matrices and character counts, for each set, were also calculated. The experiment thus evaluates six analytical strategies for each sequence set. To summarize, three levels of classification problem (easy, moderate, and hard) are examined using six strategies ten times.

The analysis began with an analysis of the variance of each matrix in order to assess the distinctiveness of sequences generated under the eight rules. Cluster analysis was then conducted on all matrices to see which produced the best reconstruction of the original eight groups. This process reveals which measurement strategy best identifies the eight generation rules.

4 Evaluation of measures of sequence similarity

The intersequence distance data generated by the similarity algorithms were subjected to hierarchical clustering to produce tree structures defining similar groups. Clusters are formed at all levels of aggregation from the initial joining of individual pairs to the combination of large groups into a final tree. The selection of a correct or meaningful set of clusters is usually part of the objective of the research. Eight clusters are built into this experiment by the generator, but the number of clusters in a research project would have to be justified in the context of the research problem.

Identification of clusters on the basis of their membership is simple when subsets in the data are distinct, as in the PROB80 sets, but becomes difficult when subsets are indistinct. The probabilistic determination of character frequencies means that sequences generated by different rules may have a similar appearance, allowing clustering algorithms to join sequences from different generation rules in the tree building process. The BCnx2, ABnx, and BAnx rules tend to produce some sequences that are similar at low PROB values. Developers of classification trees identify nodes by the characteristic that contributes most of the members and call this practice the 'plurality rule' (Breiman et al, 1984). We follow this practice, identifying a cluster with the rule that contributes the most members, but call the latter the main rule.

For the PROB80 datasets the identification of clusters in terms of the proportion of members contributed by the main rule can be considered as definitive because the sequence appearance can be reliably associated with particular generation rules. In fact, PROB = 0.80 was chosen because some alignments could cluster the datasets perfectly at that level of uniformity. However, as the classification problem becomes more difficult, a given sequence may be generated by one rule but may resemble those created by another. At PROB = 0.35, the generation of a sequence by a rule is no longer sufficient reason to believe that the sequence looks like others made by the same rule, and the assignment of the sequence to a different rule is not necessarily an error. At the lower PROB values the classification problem evolves from one of evaluating a candidate assignment against a true criterion, to a problem of measuring the agreement of independent raters.

The literature on rating agreement is extensive because the problem is central to many fields, including medical diagnostics, psychological testing, educational testing, and others (Agresti, 1996; Mackinnon, 2000). The consensus regarding testing is that simple measures of agreement between raters (or concordance) should be used, although these should be supplemented by contingency table testing methods, such as χ^2 , the odds ratio, and Cohen's kappa, as appropriate.

An 8×8 contingency table will disaggregate the membership of each identified cluster into the rules that generated the sequences. The marginal values of such a table are the rule counts and cluster sizes. Marginal totals are all 30 for the generation rules and are variable but sum to 240 for the clusters. The diagonal cells count the number of agreements between the generation rules and the cluster membership. The sum of the diagonal elements gives the total concordance, which, divided by 240, gives the proportional agreement or the proportion of the sequences from the main rule in all clusters. These ratios will be the main indices of classification success.

It should be emphasized that the objective here is to measure relative agreement rates of different algorithms, not to test the independence of ratings. If the classification procedures are effective we would expect statistical tests to reject a hypothesis of independence between generation rules and clusters. χ^2 -statistics for some contingency tables that disaggregate clusters by rules exceeded 400, which, for 49 degrees of freedom, does in fact reject a hypothesis of independence.

The mixing of appearances of sequences generated by different rules has parallels in ordinary behaviour. Diary data that record activities will contain some random content. The activities of, for example, an employed man could be the same as those of a retired woman on a particular day if unusual events occur, even though the typical patterns of these two types are vastly different. A sequence classification would, properly, include both of these patterns in one group for that day. Identification of individuals grouped in this way according to their socioeconomic characteristics is another research problem.

5 Results

5.1 PROB80, PROB50, and PROB35 similarity measures

Table 3 (over) shows the averages of the means and standard deviations of the ClustalG scores and the Gower similarities for the ten replications of the three classification problems. The table contains statistics for the whole matrix and for the submatrices of within-rule and between-rule sequence comparisons.

The 240 sequences in a set generate 28 680 unique sequence similarity comparisons. No similarity is defined for a sequence with itself. Every similarity matrix is composed of thirty-six submatrices of within-rule and between-rule comparisons. The eight within-rule submatrices are symmetrical, so the thirty sequences have 435 unique comparison pairs. These are the diagonal cells of figure A2. The twenty-eight between-rule matrices contain 30×30 combinations of interrule comparisons and thus each has 900 comparison pairs. Therefore, the observed similarities from a sample of sequences will be dominated by intergroup comparisons (if meaningful groupings can be shown to exist). Figure A2 in the appendix contains means and standard deviations of the pairwise similarity scores for the thirty-six combinations of sequence-generation rules.

The means of scores range from 108 to 185 for the PROB80 dataset and increase for the PROB50 and PROB35 datasets. Means for the whole matrices are significantly lower than any of the within-group means, indicating that the within-group similarities are higher than between-group similarities, as was intended. The Gower similarities decline as the sequences become less similar. The mean scores for the low-penalty options are very

Table 3. Descriptive statistics (PROB80, PROB50, and PROB35 sequence sets).

Classification problem	n	Statistic	Alignment similarity				Gower similarity
			local 1/0.1 ^a	local 8/3 ^a	global 1/0.1 ^a	global 8/3 ^a	
<i>Easy (PROB80)</i>							
Whole matrix	28 680	mean	185	132	183	108	8.8
		sd	74	63	74	65	5.9
Within rule	3 480	mean	316	245	315	230	19.4
		sd	37	48	38	53	6.2
Between rule	25 200	mean	167	116	165	92	7.3
		sd	58	46	58	46	4.0
<i>Moderate (PROB50)</i>							
Whole matrix	28 680	mean	226	145	224	121	8.5
		sd	29	25	29	28	2.9
Within rule	3 480	mean	267	179	264	159	11.3
		sd	18	20	19	23	3.0
Between rule	25 200	mean	221	141	219	116	8.1
		sd	25	21	25	25	2.7
<i>Hard (PROB35)</i>							
Whole matrix	28 680	mean	236	149	234	125	8.3
		sd	22	19	22	23	2.7
Within rule	3 480	mean	256	166	255	144	9.6
		sd	18	18	18	21	2.8
Between rule	25 200	mean	233	147	231	123	8.1
		sd	21	18	21	22	2.6

Note: sd = standard deviation.
^a Gap options in the form: opening penalty/extension penalty.

close in the case of both algorithms and are higher than means for the high-penalty options.

An unexpected feature of the data in table 3 is that mean similarities are higher and less dispersed for the PROB35 sequences than for the PROB50 and PROB80 sequences, which are more distinctive. The explanation is that, although more distinctive sequences have higher mean within-rule similarities, they have lower between-rule similarities. As the between-rule similarity matrices are twice as large as the within-rule matrices, and are more numerous, the lower similarity values dominate the whole matrix. The means increase by less than 5% from the PROB50 to PROB35 datasets. This suggests that, as the proportion of noise in the generation process becomes very large (more than 50%), the average similarity values stabilize.

To give some context to the descriptive statistics for the experimental dataset, we coded 391 randomly selected diaries from the Statistics Canada 1992 General Social Survey of time use into six activity categories using thirty-minute time intervals. The resulting activity sequences averaged fifty-three characters (because short episodes were rounded up), which is quite close to the length of the experimental sequences. An alignment with the local, high-penalty settings gave a mean pairwise similarity score of 260 with a standard deviation of 65. The Monte Carlo sequences thus display lower similarities than time-use diary data. The ranges of the test means and standard deviations bracket those of the real time-use data.

5.2 PROB80, PROB50, and PROB35 analysis of variance (ANOVA)

The total set of 28 680 similarities from the upper triangle of each matrix was subjected to a number of analyses of variance. The set of eight generation rules allows a number

of groupings or treatments to be defined. The most detailed is to use all thirty-six combinations of within-rule and between-rule comparisons. These are illustrated in either the upper or lower triangle of submatrices shown in figure A2. This creates a treatment variable with thirty-six values and thirty-five degrees of freedom. The conduct of these ANOVA tests takes the underlying generation processes as given or known in advance, which they are because sequences came from the Monte Carlo generator. Significant *F*-statistics will let us reject an hypothesis of equality of mean similarities among sequence sets. The tests, therefore, supply evidence as to whether or not the various generators (treatments) can be assumed to be generating the same sequences.

All alignment measures and the Gower measure were analyzed for three sequence sets (PROB80, PROB50, and PROB35). Table 4 gives the *F*-statistics for one of the PROB35 datasets. All values exceed the critical values for 0.0001 significance-level tests. It is clear that all of the alignments and the Gower measure can discriminate among the generation rules and all combinations of rules implemented here. All settings of the parameters produce *F*-statistics that reject the homogeneity of generation processes. The *F*-statistics for PROB50 datasets were higher than those for PROB35 and some PROB80 values exceeded 7000. The results indicate that ANOVA can discriminate very clearly among the sequences from different generators and hence that reproduction of the Monte Carlo generation rules is a criterion for evaluating the reliability of similarity measures.

Table 4. Analysis of variance of PROB35 similarity measures.

Similarity measure	<i>F</i> -statistic
Local 8/3 ^a	219
Local 1/0.1 ^a	293
Global 8/3 ^a	104
Global 1/0.1 ^a	264
Gower	62
Degrees of freedom = 35	
<i>n</i> = 28 644	
^a Gap options in the form: opening penalty/extension penalty.	

5.3 Identification of generation rules using cluster analysis

Table 5 (over) shows all clusters present in the first PROB80 dataset. The table identifies the clusters and gives the size and the count of sequences contributed by the main rule for each cluster. For a simple classification problem an ideal clustering will produce eight clusters of thirty sequences with all sequences from one rule in each cluster. For more complex classification problems solution quality can only be defined in terms of a more reliable procedure, as sequences made by different rules can look very much alike in terms of the number, order, and transition patterns of characters. The character-count results come from the clustering of frequency counts, not from similarity matrices. The cluster analysis perfectly grouped the L1 similarity matrix. All measures, including the Gower matrix and the character counts, were highly reliable. The G8 matrix performed most poorly, having thirty assignment errors. These results differ slightly when different sets of sequences are used and not all PROB80 datasets are perfectly classified by the L1 matrix. The anomalous result for the G8 matrix is somewhat surprising, given the conclusion that Gower similarity is a special case of the high-penalty alignment similarity. Examination of the G8 and Gower scores indicate that their product – moment correlations exceed 0.95.

Table 5. PROB80 cluster analyses.

Cluster name	Similarity matrix										Character counts	
	L1		G1		Gower		L8		G8		size	main rule count
	size	main rule count ^a	size	main rule count ^a	size	main rule count ^a	size	main rule count ^a	size	main rule count ^a		
ABC75	30	30	29	29	30	30	32	30	30	30	30	29
DEF75	30	30	30	30	31	30	29	29	58	29	29	29
ABnex	30	30	31	30	30	30	30	30	23	23	23	23
BAnex	30	30	30	30	30	30	28	28	28	28	36	29
EFnex	30	30	30	30	29	29	30	30	23	23	33	30
FEnex	30	30	30	30	30	30	31	30	25	24	27	27
BCnx2	30	30	30	30	30	30	30	30	25	25	31	30
DEnx2	30	30	30	30	30	30	30	30	28	28	31	30
Total	240	240	240	239	240	239	240	237	240	210	240	227
% of main rule	100		99.6		99.6		98.7		87.5		94.6	

^a Count of sequences contributed by the main rule for each cluster.

Because at least one option could distinguish the generation rules perfectly, the assignment of a sequence to a cluster generated by a different rule can be treated as an error. The accuracy of the six measures are thus given by their main-rule percentages.

5.4 Ten replicated tests

The concordance ratios for the eight clusters were averaged for the ten replications and the results are reported in table 6. The sum or mean of independent random variables tends to normality as the number in the sum rises and ten is a reasonably large number of replications for a Monte Carlo process (see Hammersley and Handscomb, 1964, chapter 2).

The PROB50 and PROB35 matrices cannot be evaluated in terms of absolute accuracy because no matrix allows perfect and unambiguous agreement between generation rules and the derived clusters. We can compare the six options to show which gives the clearest assignments and rate the others in comparison with the one that best replicates the known generation rules. The PROB80 agreement rates reproduce the results for the single dataset reported in table 5. The agreement rates for the PROB50 sequences average over 86% for both of the low-penalty alignments and just over 82% for the high-penalty alignments. The rate for the Gower measure is 73% and that for character counts is 65%. We note that the performance of the G8 alignment parameter set is now comparable to the L8 set and is superior to the Gower measure and character counts.

Table 6. Agreement rates for ten replications of cluster analysis of Monte Carlo sequences.

Score	Statistic	Global 1	Global 8	Local 1	Local 8	Gower	Character count
PROB80	mean	99.6	89.0	99.6	99.3	98.3	93.7
	standard error ^a	0.1	0.6	0.1	0.2	0.3	0.5
PROB50	mean	86.7	82.4	87.8	83.3	73.0	65.4
	standard error ^a	0.9	1.1	0.5	1.1	1.5	1.6
PROB35	mean	65.7	55.6	60.1	52.4	42.6	46.0
	standard error ^a	1.5	0.9	1.6	1.2	1.1	0.7

^a Standard deviations for the agreement rates are three times the standard errors.

For the more difficult PROB35 sets, the agreement rates of the global options are higher than the local options for both penalty settings. The Global-1 similarities have the highest agreements, at over 65% compared with 60.1% for the Local-1 similarities. For the high-penalty options the agreement rates are somewhat lower but the global algorithm is still more reliable. It appears that, given greater randomness, the global algorithms are more successful in finding weak signals and making correct assignments.

The Gower similarity and character counting gave excellent agreement for the simpler PROB80 classification problem, but their agreement rates declined substantially for the PROB35 sets.

A purely random assignment of 240 objects to eight clusters would be expected to produce thirty correct assignments or 12.5% agreement. Thus, for the PROB35 sequences, the low-penalty alignments may be interpreted as operating over five times more effectively than random assignments. The high-penalty algorithms operate between four and five times more effectively. The Gower and character counting measures are over three times more effective than random assignment.

The results indicate that the clearest, least-ambiguous classification strategy is to align with low gap penalties. This does not discriminate clusters perfectly even for the

PROB80 datasets, although here it is highly reliable, exceeding 99%. All strategies display lower agreement with the generator as the difficulty of the classification problem increases, as in datasets PROB50 and PROB35.

6 Strategies for alignment analysis of event sequences

6.1 ANOVA results

In most applications there will be no preliminary ANOVA of group means because, before the analysis, the investigator will not have identified behavioural groupings. The ANOVA tests were undertaken to assess the distinctiveness of groups of sequences generated by different rules. In this experiment, all gap penalties and algorithm options produced alignments that discriminated clearly among generation rules for distinctive datasets (PROB80) and for those produced by much less distinctive rules. The generator is working as intended and there are eight reasonably distinct groups for the software to find.

6.2 Extraction of clusters

The most successful alignment strategy for identifying generation rules through the use of all test datasets was to use very low gap penalties. The local alignment option performed slightly better than the global option for easier classification problems, but the global option was better with the more difficult PROB35 dataset.

The reversal of effectiveness of local and global algorithms is difficult to explain. The local algorithm may be effective in classifying highly similar sequences because it gathers enough information for good classification in high-similarity regions and can disregard contradictory data from dissimilar regions. The global algorithm is obliged to score both complete sequences and the scores may thus be watered down by data from dissimilar regions. As the information content of sequences is reduced in more difficult classification problems, the global algorithm may simply gather more information by considering all positions.

Most social-science alignment applications have used the global algorithm with relatively high gap penalties. The conclusion from this experiment is that the selection of algorithm is not critical. However, in cases in which the objective is classification, the similarity and distance measures of event sequences that employ very low gap penalties can be clustered into clearer behavioural patterns.

An unexpected result that emerged from the analysis was that average sample similarities are higher among sequences with weaker patterns than those with stronger patterns. Strong patterns mean that intergroup similarities are low and that there are more possible intergroup comparisons in a sample than within-group comparisons. Researchers may find highly distinctive behavioural groupings in samples with relatively low average similarity measures. Conversely, samples with relatively high average similarity may contain groups with indistinct behavioural patterns. It is no surprise that the examination of heterogeneous groups will be more interesting than the examination of homogeneous groups. Dissecting a pea pod is more interesting than dissecting a turnip. Consideration of average similarity values points the researcher to populations with lower, rather than higher, scores.

In real research situations the analyst will not normally know how many, if any, patterns are present in the data because finding such patterns is probably the object of the research. There would then be no justification for picking a certain number of clusters to find. A possible strategy would be to deliberately set the number of clusters to be fairly high—say, the square root of the sample size. In this case that would be about 16. Groups identified in this way would generally be more homogeneous than the larger final clusters extracted in this experiment. By stopping the clustering procedure

early, analysts can determine for themselves the best way to put the building blocks together.

6.3 Identification of clusters

In real research projects neither the similarity matrices nor the log and tree files will have labels that identify the behavioural group that created the event sequence. The investigator will see only clusters of individuals that share a sequential pattern and will have to investigate the characteristics of the group members to identify the membership. This can be done by using some kind of multivariate analysis. ANOVA has been used to identify the socioeconomic characteristics of respondents to time-use surveys (Wilson, 2001). Logistic regression has the additional advantage that it indicates the relative contribution of the independent factors in the assignment of individuals to behavioural groups.

6.4 Appearance of aligned sequences

After classifying even sequences into some set of groups, an analyst may want to see the patterns. This is a case in which multiple alignments are useful. However, multiple alignments based on exceptionally low gap penalties are not recognizable as representations of the raw data, either synthetic or real. Higher gap penalties produce alignments that are truer to life than low penalties.

A difficult and as yet largely ignored question relating to alignment applications outside of molecular biology is that of how to interpret indels, as they do not exist in the data and are an artifact of the algorithm. In biology gaps can sometimes be interpreted as having been created by mutations, thereby illustrating something of the evolutionary history of proteins or nucleotides. In the field of transportation planning, models of activity scheduling being developed by Joh et al (2002) and by Doherty and Miller (2000) propose that people actually insert and delete activities into and from planned agendas from hour to hour during the day. Surprise events may cause plans or routines to be rewritten 'on the fly'. This suggests that regions with gaps may be illustrative of activity patterns that have been amended to accommodate unexpected events. In cases in which sequences have substantially different lengths, as in the case of diary surveys, in which people simply describe their activities in greater or less detail, indels, especially at the ends of aligned sequences, may not have any positive or negative implications. However, if sequence data are naturally of a fixed length, such as diaries for equal time intervals, the analyst may try to minimize the number of indels used in the alignment.

6.5 The best strategy

If examining data coded as character sequences, the key question is whether a classification is being produced, or whether the analysis is to include a visual representation of data that should remain faithful to the real events. On the basis of this study the strongest classifications were based on very low gap penalties that would not produce anything like real representations of the data. Such settings allow the algorithms to find as many exact and near matches as possible. Group membership is thus determined by the most sensitive parameters. At a later step, group membership can be fixed and multiple alignment parameters can be set higher to force the alignment file to illustrate more realistic and interpretable event patterns. Last, socioeconomic characteristics of groups can be analysed by logistic regression or some other multivariate method.

A mix of sequences from different rules is not interpreted as a classification error unless we have reason to believe that some method could have classified the groups perfectly, as in the case of the PROB80 dataset. Rather, it is argued that different

generation processes with some random elements simply do not produce highly distinct character sequences. The agreement rates quoted here are interpreted as the lower bounds of the classification reliability. The true assessment of classification reliability can be made only in terms of a better classifier. To date, for broad classes of sequential social events, the most reliable classifiers are alignment algorithms.

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Appendix

> # 1 ADBBEBDDAAADAABAACBCFBABFCFCDACBFECAFBBBACECBCBBCC
> # 2 CEDBABDDCABBAACEECDCBFDCBBCFDAABBBAFECBAFBDBBCAB
> # 3 AAFADADDCCFDEFDFEAEBEDEFDFDCFFBBEDCFEEFCFCEDAFBDE
> # 4 FFEDDDFFDEEDDEDFCDCDBCCFFBFBDBDCEFDEDFDEBDDFFDFDCFF
> # 5 AAFAAFADABAACBBAAABFBFBABABABBBBBBBBBBBBBBDBBBEBA
> # 6 EEADAFACAAAAFABABADBAAEAFABBBAAAABBBABDBBBBFBFB
> # 7 BBBBBDDBABBADEBBABFAAFFFDABDDBAABAAEAAAAAAAAADABF
> # 8 BBBBFBABBBABCBABAAADADABDAFFABABAAAAAAAAAFAABDE
> # 9 ECEEEFBFEEEEEEEFCEFAEEFEFFEEFFFAEFFDEFBFAFFCFFFFF
> #10 EEDCEDEEEFEFCFFFEFFFAFCFEFFCFEFFFFFEDDFFFFAFFEF
> #11 FFFFEFFFEFFFEAEDCFEFAEBDDFFAEFBEEEFEEEEEBBEEAE
> #12 FFFFAFFFBEECAEFEEFFEEDEAFAFBEECFEEEEEEEEEEAAEEAD
> #13 CCBBCBBBCBBBCCCCBCCCCDCFCDCCCCCCFEFCFDCC
> #14 BCCCCBBBEBBBBBBFEFCABBBBCBCCCCDFCEACFCCCCACCFFE
> #15 DDEDEEEDDBDDCDBDDDDDECEDECEDEEAFBCEEFEEEEEEEEEC
> #16 DEDEDEDDDDDDDDDDDDDDCEFEFEDEEEEEEEEEEDEFEEEEEEEB

Note: for input to ClustalG, sequence labels (>#1 etc.) are on a line preceding the sequence data.

Figure A1. Sample sequences from eight generation rules (PROB80 dataset).

PROB80									
	ABC75	DEF75	ABnex	BAnex	EFnex	FEnex	BCnx2	DEnx2	
PROB50	ABC75 Mean Sd Range	189 18.4 130–250	127 17.1 78–190	180 19.7 120–230	180 21.4 129–240	101 18.8 50–170	99 16.9 58–149	163 20.9 100–203	92 18.7 40–150
	DEF75 Mean Sd Range	127 17.1 78–190	186 15.7 140–239	98 20.6 39–160	99 19.0 40–159	172 20.9 119–230	172 22.3 118–250	103 19.5 50–150	168 19.3 120–220
	ABnex Mean Sd Range	172 17.7 129–240	142 18.7 87–199	266/184 22.9/17.1 209–340/ 140–230	183 19.6 130–260	75 18.3 29–150	77 19.5 30–160	155 17.5 109–209	74 20.2 20–130
	BAnex Mean Sd Range	174 18.7 119–240	143 18.1 69–209	154 16.8 109–210	276/190 19.9/17.1 219–359/ 149–240	74 18.4 20–149	75 17.4 20–139	136 18.3 80–199	70 16.7 20–119
	EFnex Mean Sd Range	145 18.4 90–219	168 18.5 119–229	147 19.5 99–210	150 19.7 90–210	273/193 23.3/19.8 200–330/ 140–240	181 22.4 110–249	77 18.2 40–160	138 19.4 90–200
	FEnex Mean Sd Range	135 19.3 79–210	169 19.0 120–240	147 20.1 89–210	144 18.0 89–210	146 16.7 100–209	266/194 22.3/22.0 200–330/ 140–259	74 17.9 30–139	197 25.1 120–279
	BCnx2 Mean Sd Range	164 17.5 69–220	143 16.6 89–200	152 17.3 109–209	159 18.8 110–219	148 20.2 80–230	149 19.2 99–220	293/202 21.3/20.7 239–350/ 149–270	73 17.9 30–130
	DEnx2 Mean Sd Range	138 19.5 70–207	167 17.7 118–229	149 19.1 89–220	146 18.1 98–210	146 17.7 99–210	176 21.4 120–259	145 19.1 80–209	291/199 24.5/21.1 230–370/ 140–270

Note: lower triangle contains PROB50 interruler similarity means, standard deviations, and ranges. Upper triangle contains PROB80 data. Diagonal cells contain intraruler data in the form PROB80 data/PROB50 data. ABC75 and DEF75 sequences were the same in both datasets. Sd, standard deviation.

Figure A2. Local alignment algorithm, opening penalty = 1, extension penalty = 0.1. Alignment score means, standard deviations, and ranges of all combinations of generation rules.

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